

# Experiment 'p\_3\_1\_classify\_int' Results

June 4, 2026

**Experiment outcome:** SUCCESS

**Bad responses:** 0

**Responses containing assume :** 0

**Responses containing expect :** 0

**Resolution attempts:** 1

**Hard fails (resolution):** 0

**Soft fails (resolution):** 0

**Verification attempts:** 1

## Problem Specification

**Problem name:** p\_3\_1\_classify\_int

**Natural language statement:** null

**Method signature:** p\_3\_1\_classify\_int (x: int) returns (s: string)

## Ensures

- $x == 0 \implies s == \text{"zero"}$
- $x > 0 \implies s == \text{"positive"}$
- $x < 0 \implies s == \text{"negative"}$

## GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

### Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_3_1_classify_int (x: int) returns (s: string)`

The method should respect the following contract:

`ensures x == 0 ==> s == "zero", ensures x > 0 ==> s == "positive", ensures x < 0 ==> s ==  
↪ "negative"`

Produce and show only the Dafny body of this method, including the curly braces that  
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor  
↪ 'expect' anywhere in your code.

### GenAI → Program

**System fingerprint:** null

**ID:** chatcmpl-Dn33zHRB192XftOPpFZBGTCh4vv8x

**Model:** gpt-5.5-2026-04-23

**Created at:** 1780582219

**Finish reason:** stop

**Usage:** 300 tokens in, and 149 tokens out

```
{  
  if x == 0 {  
    s := "zero";  
  } else if x > 0 {  
    s := "positive";  
  } else {  
    s := "negative";  
  }  
}
```

## Final Program

```
method p_3_1_classify_int (x: int) returns (s: string)  
  ensures x == 0 ==> s == "zero"  
  ensures x > 0 ==> s == "positive"  
  ensures x < 0 ==> s == "negative"  
{  
  if x == 0 {
```

```
    s := "zero";  
  } else if x > 0 {  
    s := "positive";  
  } else {  
    s := "negative";  
  }  
}
```

## Total Token Usage

Input tokens: 300

Output tokens: 149

Reasoning tokens: 79

Sum of 'total tokens': 449

## Experiment Timings

**Overall Experiment** started at 1780582205146, ended at 1780582267137, lasting 61991ms (61.99 seconds)

**Iteration #1** started at 1780582205399, ended at 1780582267135, lasting 61736ms (61.74 seconds)